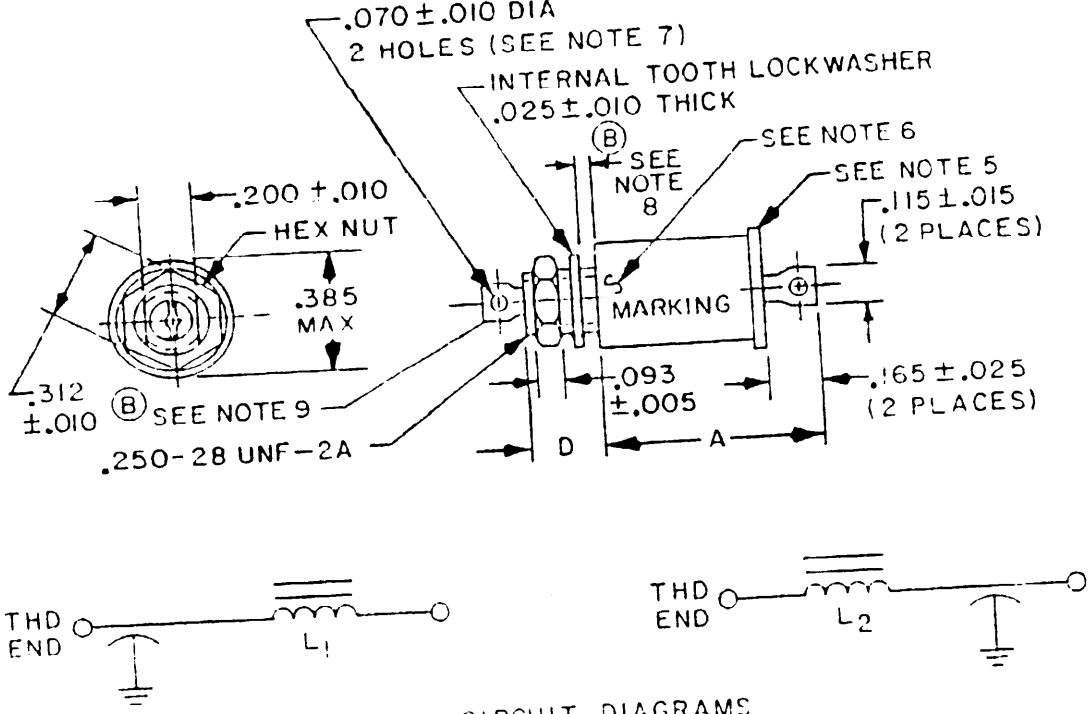
MILITARY SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE,
HERMETICALLY SEALED, STYLE FL12

Ⓑ Inactive for new design after 2 September 1988.
For new design use MIL-F-15733/58 or MIL-F-28861/4.
See table II for substitution data.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the the solicitation: MIL-F-15733.



CIRCUIT DIAGRAMS

FIGURE 1. Case dimensions and circuit diagrams.

Ⓑ denotes changes

AMSC N/A
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Dash number	Dimensions	
	A, maximum	D, ±.010
0002, 0004, 0008, 0013, 0014, 0021, 0022	.740	.312
0003, 0007, 0017, 0018	.740	.187
0006, 0015, 0016, 0023, 0024	.760	.312
0001, 0005, 0009, 0010, 0011, 0012, 0019, 0020	.760	.187

Inches	mm
.005	0.13
.010	0.25
.015	0.38
.025	0.63
.070	1.78
.093	2.36
.115	2.92
.165	4.19
.187	4.75
.200	5.08
.250	6.35
.312	7.93
.385	9.78
.740	18.80
.760	19.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware.
5. Use of style FL12 with or without shoulder is optional.
6. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol "C" or the symbol "L", as follows:

Circuit	Symbol
L_1	C
L_2	L

7. Optional terminal hole of .050 ±.010 X .070 ±.010 (1.27 ±0.25 mm X 1.78 ±0.25 mm) may be supplied.
8. Imperfect thread or undercut optional .050 (1.27 mm) inch maximum.
9. Terminals shall be aligned within 10° of the vertical (threaded bushing flats).

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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TABLE I. Electrical characteristics (FL12).

Dash numbers	Circuit diagram	Max rated current (amperes)	Max voltage drop (volts)	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/																	
				At +25°C						At -55°C						At +125°C					
				30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	1 GHz	30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	1 GHz	30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	1 GHz
0017, 0021	L_1	.06	4.2	44	58	70	70	70	70	33	56	58	70	70	70	34	56	58	70	70	70
0009, 0013	L_2	.06	4.2	44	68	70	"	"	"	33	56	58	70	"	"	34	56	58	70	"	"
0018, 0022	L_1	.15	1.8	24	48	58	"	"	"	12	36	46	63	"	"	12	36	46	63	"	"
0010, 0014	L_2	.15	1.8	24	48	58	"	"	"	12	36	46	63	"	"	12	36	46	63	"	"
0019, 0023	L_1	.25	1.0	18	42	51	"	"	"	6	30	39	54	"	"	6	30	39	54	"	"
0011, 0015	L_2	.25	1.0	18	42	51	"	"	"	6	30	39	54	"	"	6	30	39	54	"	"
0003, 0004	L_1	.30	.69	16	34	44	62	"	"	4	22	32	50	"	"	4	22	32	50	"	"
0001, 0002	L_2	.30	.69	16	34	44	62	"	"	4	22	32	50	"	"	4	22	32	50	"	"
0020, 0024	L_1	.45	.54	15	29	33	46	66	"	3	17	21	34	54	"	3	17	21	34	54	"
0012, 0016	L_2	.45	.54	15	29	33	46	66	"	3	17	21	34	54	"	3	17	21	34	54	"
0007, 0008	L_1	10.0	.10	15	28	31	42	56	"	3	16	19	30	44	"	3	16	19	30	44	"
0005, 0006	L_2	10.0	.10	15	28	31	42	56	"	3	16	19	30	44	"	3	16	19	30	44	"

1/ Insertion loss measurements shall be made under full load over the frequency range of 150 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no-load.

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REQUIREMENTS:

Dimensions and configurations: See figure 1 and table I.

Case: Metal.

Terminals: Solder lug.

Operating temperature range: -55°C to +125°C; temperature characteristic D

Rated voltage: 28 volts, dc (derated from 50 volts, dc at 85°C).

Rated current: See table I.

Insertion loss characteristic: Symbol W (see table I).

Creepage and clearance: Not applicable.

Seal: Method 112, test condition A, MIL-STD-202.

Capacitance to ground: Not applicable.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.

Test temperature: 25°C.

Test voltage: 2.5 times rated voltage, applied between each terminal and ground, for a period of 1 to 5 seconds.

Maximum charging current: 50 mA.

Barometric pressure (reduced): Method 105, test condition D (100,000 ft.), MIL-STD-202.

Dielectric withstanding voltage: 125 percent of the rated voltage.

Maximum charging current: 50 mA.

Insulation resistance: Method 302, MIL-STD-202.

Test temperature: 25°C.

Test potential: Rated dc voltage.

Maximum charging current: 50 mA.

Insulation resistance shall be not less than 100 megohms.

Voltage drop: See table I.

Insertion loss: In accordance with MIL-STD-220 and table I herein.

Insertion loss (check test): In accordance with MIL-F-15733.

Overload: Filters shall be suspended by their terminals in free air at an ambient temperature of 25°C ±5°C. A test current of 140 percent of rated current shall be applied for 15 minutes. After the filter has returned to room temperature, insulation resistance and voltage drop measurements shall meet initial requirements.

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Terminal strength: Method 211, test condition A (pull), MIL-STD-202.

Force: 5 pounds.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:

Thermal shock: Method 107, test condition A, except the temperature at step 3 shall be +125°C.

Immersion: Method 104, test condition A, MIL-STD-202.

Measurements after final cycling: Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Maximum charging current: 50 mA.

Insulation resistance shall be not less than 30 percent of the initial requirement.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Electrical load: 100 percent of the rated voltage between terminals and case.

Vibration, high frequency: Method 204, test condition D (20 G), MIL-STD-202.

Electrical load: Rated voltage, and current as specified in table II, shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent open or short-circuiting.

After the vibration test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the temperature at step 3 shall be +125°C. No measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.

Initial measurements: Not applicable.

Polarization voltage: 28 V dc.

Measurements after 24-hour drying period: Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Maximum charging current: 50 mA.

Insulation resistance shall be not less than 50 megohms.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

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Life: Method 108, MIL-STD-202. Test condition D (1,000 hours) for qualification inspection; test condition B (250 hours) for group C inspection. Test temperature +125°C.

Test voltage: 1.4 times rated dc voltage.

Measurements after test: Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Maximum charging current: 50 mA.

Insulation resistance shall be not less than 30 percent of the initial requirement.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

(B) Part or Identifying Number (PIN): Consists of the basic number of this specification sheet and a dash number from table I.

Example:



(B) Supersession data: See table II.

(B) TABLE II. Supersession data.

Inactive PIN MIL-F-15733/24-	Superseding specification sheet	Superseding PIN
0001	MIL-F-28861/4	M28861/04-005TB
0002	"	M28861/04-023TB
0003	"	M28861/04-004TB
0004	"	M28861/04-022TB
0005	MIL-F-15733/58	M15733/58-0001
0006	"	M15733/58-0001
0007	"	M15733/58-0002
0008	"	M15733/58-0004
0009	MIL-F-28861/4	M28861/04-002TB
0010	"	M28861/04-002TB
0011	"	M28861/04-005TB
0012	"	M28861/04-008TB
0013	"	M28861/04-020TB
0014	"	M28861/04-020TB
0015	"	M28861/04-023TB
0016	"	M28861/04-026TB
0017	"	M28861/04-001TB
0018	"	M28861/04-001TB
0019	"	M28861/04-004TB
0020	"	M28861/04-007TB
0021	"	M28861/04-019TB
0022	"	M28861/04-019TB
0023	"	M28861/04-022TB
0024	"	M28861/04-025TB

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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